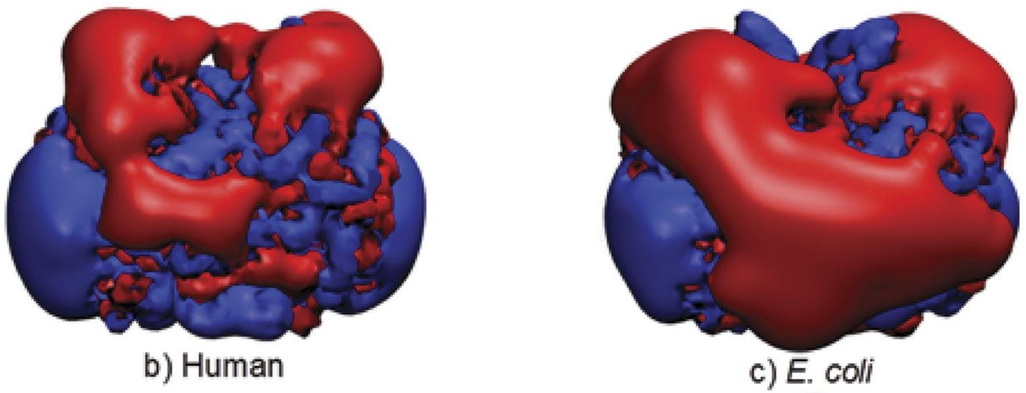
2/9 Biomolecular Electrostatics

- This week's lecture and exercise are intended to help you achieve the following learning objectives:
  - Predict the pKa's of titratable amino acids on a protein and protonation at a given pH. Summarize the qualitative relationship between electrostatic potential and pKa.
  - Compute and visualize the electrostatic potential of a protein. Compare the advantages and disadvantages of several representations.
- At the end of this lecture, you should be able to explain, in a general sense, the
  - importance of electrostatics in biological macromolecules
  - origin of the Poisson-Boltzmann equation

Roles of Biomolecular Electrostatics

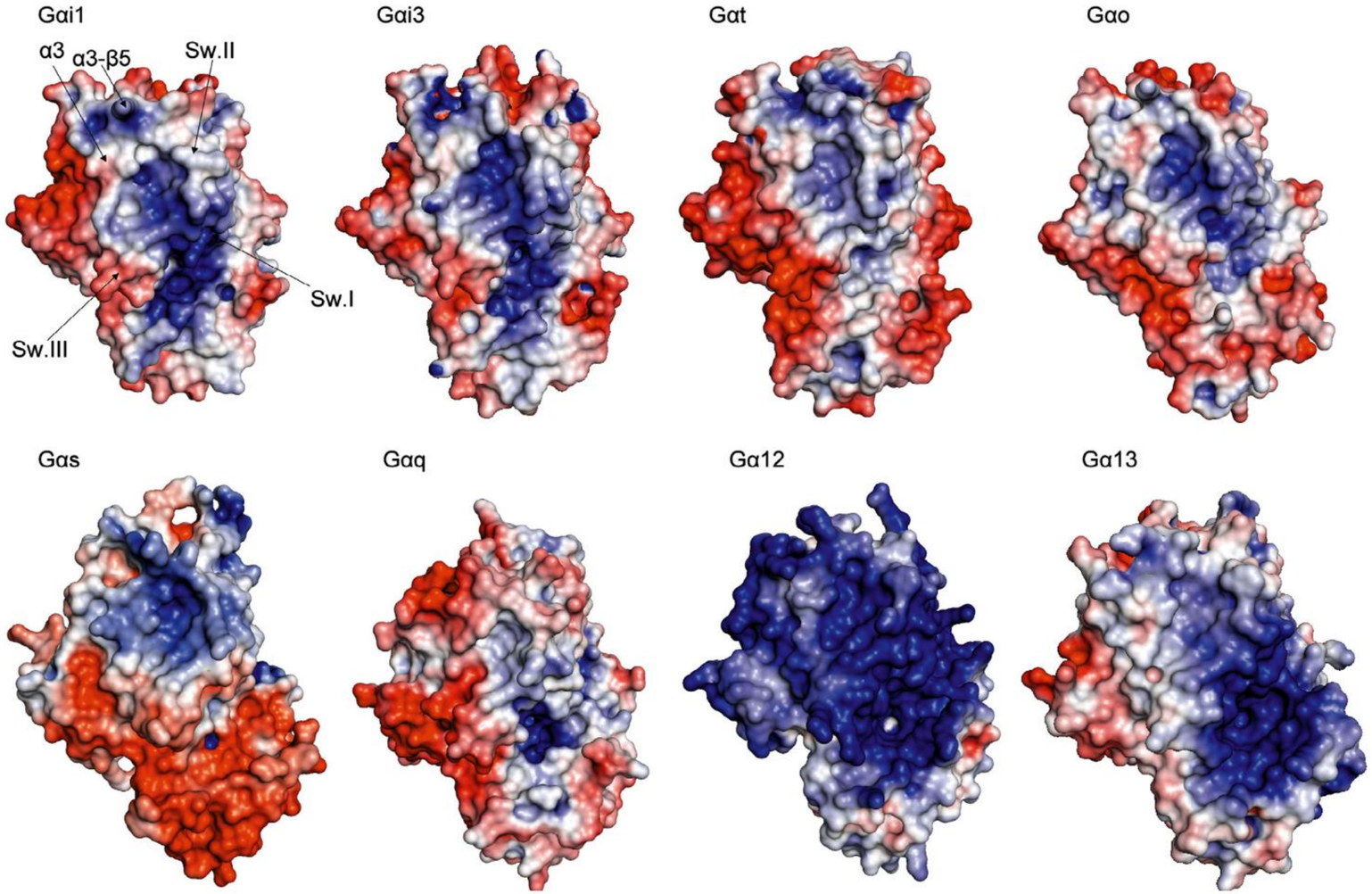
- Electrostatics important in (at least)
  - protonation, influencing side chain pKa
  - binding for
    - steering, facilitating approach of species
    - complexation, as complementarity means lower potential energy
  - enzyme catalysis, as electric potential stabilizes transition state
  - solvation
- Electrostatic potential usually conserved near functional sites

Thymidylate Synthase



blue isocontours +1 kT/e,
red isocontours -1 kT/e
From Fig 1 of Garg *et al*. (2015).

G proteins have distinct fingerprints



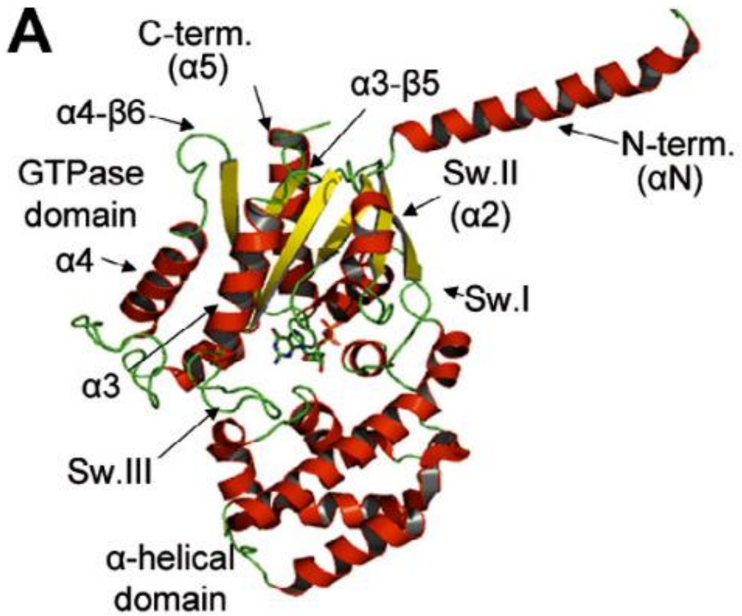


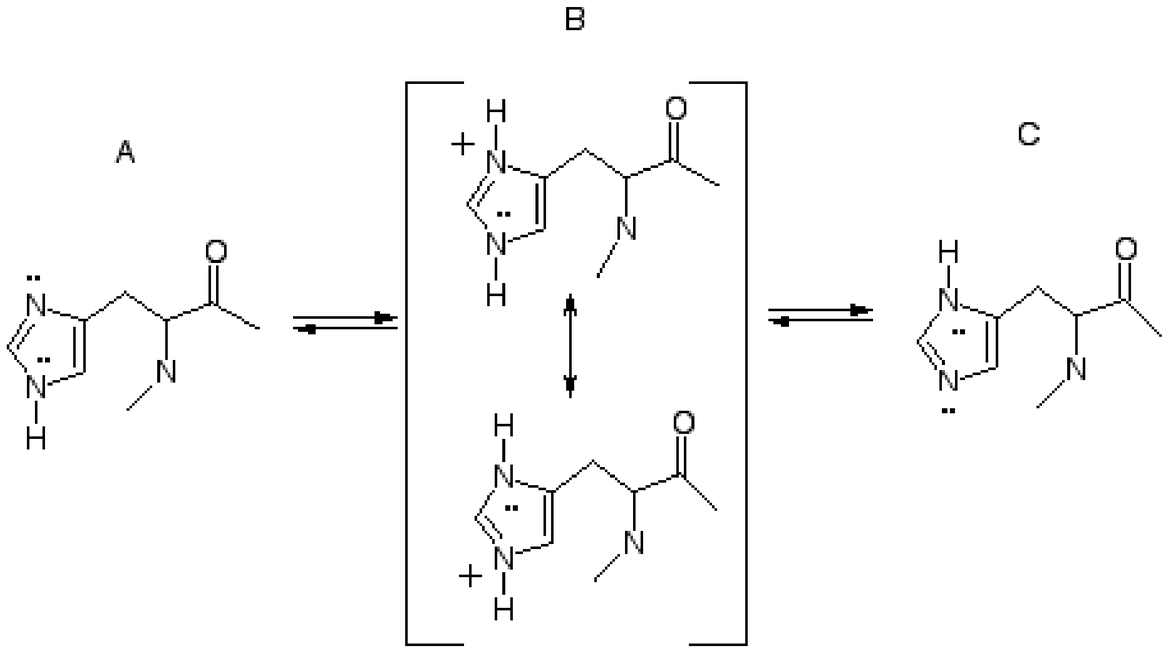
Fig.2. Electrostatic molecular surfaces of Gα subunits. All subunits are oriented in the same way as in Fig.1. Charged surfaces are colored in shades of blue for positive and red for negative charges, while uncharged surfaces are colored white. Subunit surfaces are contoured from −5 (red) to +5 (blue) kT/e⁻ based on the potential of the solvent accessible surface. All subunits are in their activated state, with the exception of $G\alpha_{i3}$ and $G\alpha_o$, which are in their RGS-bound state. There is a conserved, effector-binding pocket between Switch II and α3. A visualization of the potential of the opposite side of the Gα subunits is shown in Supplementary Fig. S6. Crystal structures used are $G\alpha_{i1}$ (PDB: 1GIA), $G\alpha_{i3}$ (PDB: 2V4Z), $G\alpha_t$ (PDB: 1TND), $G\alpha_o$ (PDB: 3C7K), $G\alpha_s$ (PDB: 1AZT), $G\alpha_q$ (PDB: 3AH8), $G\alpha_{12}$ (PDB: 1ZCA) and $G\alpha_{13}$ (PDB: 3CX8).

Baltoumas, F. A.; Theodoropoulou, M. C.; Hamodrakas, S. J. Interactions of the α-Subunits of Heterotrimeric G-Proteins with GPCRs, Effectors and RGS Proteins: A Critical Review and Analysis of Interacting Surfaces, Conformational Shifts, Structural Diversity and Electrostatic Potentials. Journal of Structural Biology 2013, 182 (3), 209–218. https://doi.org/10.1016/j.jsb.2013.03.004.

Protein Protonation

- Which structure determination technique(s) have been used to identify the protonation of amino acid side chains?
  A. X-ray crystallography
  B. X-ray crystallography, at very high resolution
  C. Nuclear magnetic resonance
  D. Cryo-electron microscopy
  E. None of the above

Histidine



https://spdbv.unil.ch/TheMolecularLevel/Goodies/Get2NoHistidine.html

- As the side chain has a pK_a ~ 7.0, imidazolium ion (B) and imidazole (A or C) are all present in physiological conditions
- Which state(s) are stabilized by a negative electrostatic potential?
- What can stabilize a specific form of imidazole?

Electrostatics is important in solvation

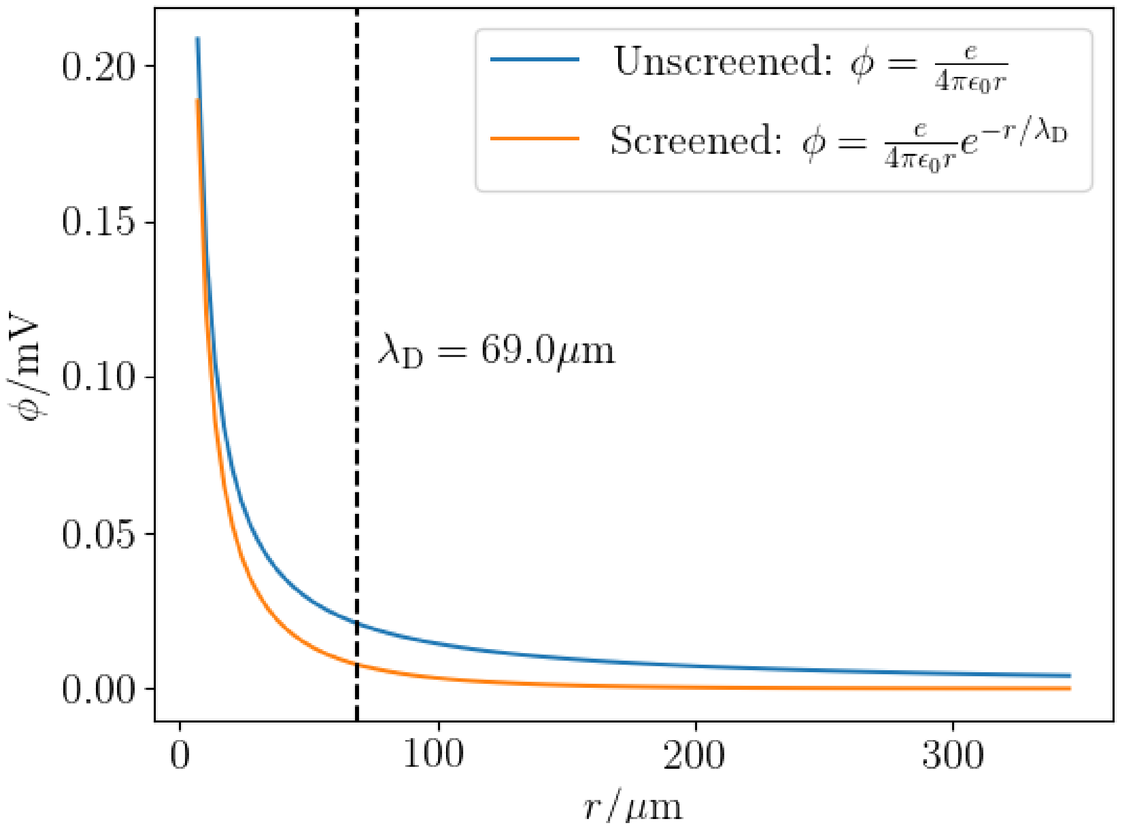
- It is an *implicit solvent* model that does not account for specific water positions
- Solvent is assumed to modify the dielectric constant
  - Coulomb's law is $F = \frac{1}{4\pi\epsilon}\frac{Q_1 Q_2}{r^2}$.
  - The permittivity ε is related to the permittivity of free space ε_0 by the dielectric constant κ, $\varepsilon = \kappa\varepsilon_0$.
    - ~80 for water
    - ~1 for the protein interior
  - The electrostatic component of the solvation energy is modeled as the difference between the energy in water and in vacuum

Modeling Electrostatics

- In biological macromolecules, the electrostatic potential is usually calculated based on the Poisson-Boltzmann equation
  - The Poisson equation $\nabla \cdot \epsilon(r)\nabla\phi(r) + 4\pi\rho(r) = 0$ describes the potential $\phi(r)$ due to a given charge density $\rho(r)$ and dielectric $\epsilon(r)$. Atoms in the biomolecule are assumed to have a fixed charge.
  - The Poisson-Boltzmann equation assumes that (infinitely small) ions surround a biomolecule in accordance with the Boltzmann distribution
- The PB equation is a partial differential equation that is solved numerically
- The equation is often linearized to be more numerically stable

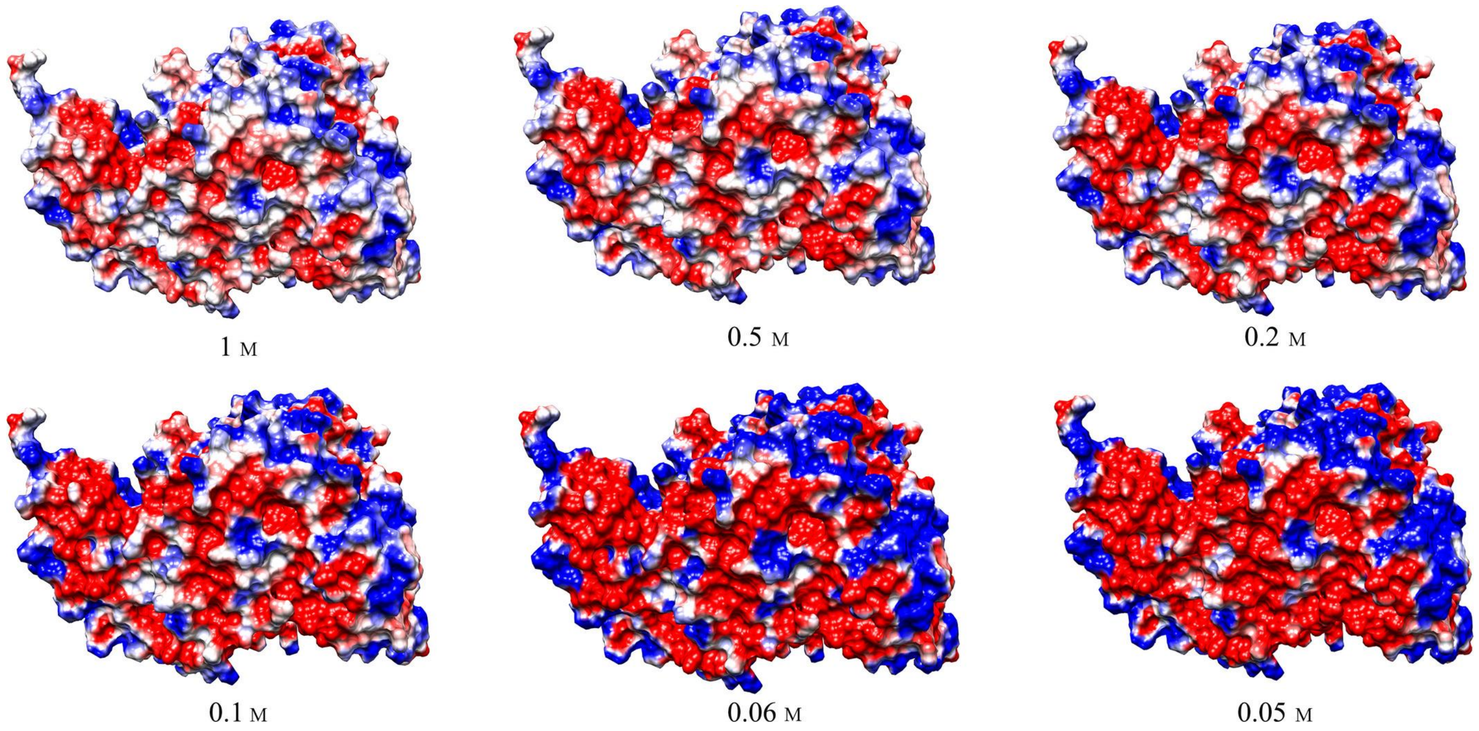
Electrostatic Screening

- Ions “screen” the electrostatics between two charges



https://scipython.com/blog/the-debye-length/

Ion concentration effects



Potential contours (−0.5 (red) and +0.5 (blue) *kT*/*e*) around AGT-Ma at pH 7.4, 37 °C, at the indicated ionic strength.
from Dindo et al (2017) https://febs.onlinelibrary.wiley.com/doi/full/10.1111/febs.14269

- What happens to the electrostatic potential as the ionic concentration is increased. Why?

Influences on biomolecular electrostatics

- Which are influences on biomolecular electrostatics?
  A. pH
  B. Concentration of ions
  C. Valency of ions (+1 vs. +2)
  D. Identity and size of ions (Ca^{2+} vs Mg^{2+})
- Which are accounted for by the Poisson-Boltzmann equation?
  A. pH
  B. Concentration of ions
  C. Valency of ions (+1 vs. +2)
  D. Identity and size of ions (Ca^{2+} vs Mg^{2+})

Review Questions

- Discuss some of the ways in which electrostatics is important for the function of biological macromolecules
- An equation often used to model the electrostatics of biological macromolecules is the Poisson-Boltzmann equation. Why does the equation have this name?
- How can the Poisson-Boltzmann equation be used to calculate the electrostatic component of the solvation free energy?

References

- Garg, D.; Skouloubris, S.; Briffotaux, J.; Myllykallio, H.; Wade, R. C. Conservation and Role of Electrostatics in Thymidylate Synthase. Sci Rep 2015, 5 (1), 17356. https://doi.org/10.1038/srep17356, adapted under the CC BY 4.0 license.
- Dindo et al (2017) https://febs.onlinelibrary.wiley.com/doi/full/10.1111/febs.14269
- Baltoumas, F. A.; Theodoropoulou, M. C.; Hamodrakas, S. J. Interactions of the α-Subunits of Heterotrimeric G-Proteins with GPCRs, Effectors and RGS Proteins: A Critical Review and Analysis of Interacting Surfaces, Conformational Shifts, Structural Diversity and Electrostatic Potentials. Journal of Structural Biology 2013, 182 (3), 209–218. https://doi.org/10.1016/j.jsb.2013.03.004.